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Research Article

Dominant HA motif 131/132-NT shapes the phenotype of avian H9N2 influenza virus by modulating agglutination property, receptor specificity and infectivity in animals

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ABSTRACT

H9N2 avian influenza viruses (AIVs), which are enzootic in poultry and possess spillover potential to humans, represent a considerable global public health threat. Since 2021, we have observed an unprecedented decline in the ability of H9N2 viruses to agglutinate chicken red blood cells (CRBCs), a change that impairs effective detection in routine surveillance. Substituting CRBCs with turkey or guinea pig erythrocytes in hemagglutination assays effectively restored virus detectability. Through integrated bioinformatic and biological approaches, we identified that this phenotype is associated with specific amino acid substitutions at residues 131 and 132 of the 130-loop in the hemagglutinin (HA) protein. The HA motif 131/132-NT emerged in 2015 and rapidly increased in frequency, becoming the dominant pattern since 2021 in chickens and humans. Critically, the 131/132-NT motif attenuated viral binding affinity to short-chain α2,6-linked sialic acid receptors, explaining the loss of agglutination with CRBCs. Animal experiments further demonstrated that viruses carrying the HA 131/132-NT mutations maintained strong infectivity in chickens but altered tissue tropism in mice, showing reduced replication in the lungs. Collectively, our findings reveal a potential surveillance gap caused by reduced hemagglutination activity of H9N2 viruses, which may compromise the sensitivity of CRBC-based detection and potentially lead to under-detection of circulating viruses. Incorporating turkey or guinea pig red blood cells into surveillance protocols is therefore recommended to enhance detection sensitivity.

INTRODUCTION

The H9N2 subtype avian influenza virus (AIV) is a globally prevalent pathogen that has been detected in poultry and wild birds across more than 60 countries (Bi et al., 2022). Despite its generally low pathogenicity in avian species, the H9N2 virus serves as an important gene donor, contributing internal gene segments to zoonotic influenza viruses, including H5N6, H7N9, H10N3, and H3N8, thereby increasing the risk of cross-species transmission (Gu et al., 2014; Pu et al., 2015; Shen et al., 2016; Qi et al., 2022; Yang et al., 2022). As of November 25,

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2025, a total of 182 human infections with H9N2 virus have been reported worldwide, 114 of which occurred between 2021 and 2025 (Buczkowski et al., 2025), posing an increasing threat to both poultry and public health.

The hemagglutination assay, a fundamental tool for influenza surveillance, is based on the binding of viral hemagglutinin (HA) to sialic acid (SA) receptors on red blood cells (Hirst, 1941). The receptor-binding property of HA critically determines viral entry efficiency and cross-species transmission potential (Imai and Kawaoka, 2012). Under host or vaccine pressures, mutations in the HA receptor-binding domain (RBD) can alter receptor-binding properties, thereby influencing hemagglutination behavior (Ge and Wang, 2011; Long et al., 2019). Previous studies have shown that receptor-binding evolution in other influenza subtypes, such as human H3N2 viruses, can be accompanied by loss of chicken red blood cell agglutination. Generally, human influenza viruses and AIVs preferentially bind to cell surface glycans carrying terminal α2,6-linked and α2,3-linked sialic acid moieties, respectively (Matrosovich et al., 1997; Matrosovich et al., 2000). However, most H9N2 AIVs have acquired the ability to bind human-like α2,6-linked receptors (Wan and Perez, 2007; Zou et al., 2019; Bi et al., 2020). In our ongoing surveillance, we observed that recent H9N2 isolates displayed altered hemagglutination capacity with chicken red blood cells (CRBCs), suggesting emerging changes in their receptor-binding characteristics. However, the molecular basis and biological implications of the altered hemagglutination phenotype observed in H9N2 viruses remain unclear.

In this study, we systematically analyzed the hemagglutination of H9N2 strains isolated in China (1998–2023). We found that, starting from 2021, the H9N2 strains with reduced hemagglutination activity inCRBCs have become the predominant circulating strains, while maintaining normal activity with turkey red blood cells (TRBCs) and guinea pig red blood cells (GRBCs). Mechanistic studies revealed that HA 130-loop mutations at residues 131/132-NT weakened viral binding to short-chain α2,6-linked sialic acid receptors, thereby reducing hemagglutination with CRBCs while maintaining strong infectivity in chickens. Notably, this motif also altered viral tissue tropism in mice, favoring replication in the upper respiratory tract and potentially increasing the risk of cross-species adaptation. These findings highlight changes in the hemagglutination characteristics of H9N2 viruses since 2021, and provide valuable insight for improved surveillance and control strategies.

RESULTS

H9N2 AIV strains (2021–2023) exhibited reduced hemagglutination of chicken RBCs but retained agglutination of Turkey and Guinea pig RBCs

Recently, our ongoing surveillance revealed that the H9N2 AIV strains isolated since 2021 exhibit reduced hemagglutination of CRBCs. To explore the evolutionary context of this phenomenon, phylogenetic analysis of *HA* genes from the GISAID database and 25 isolates

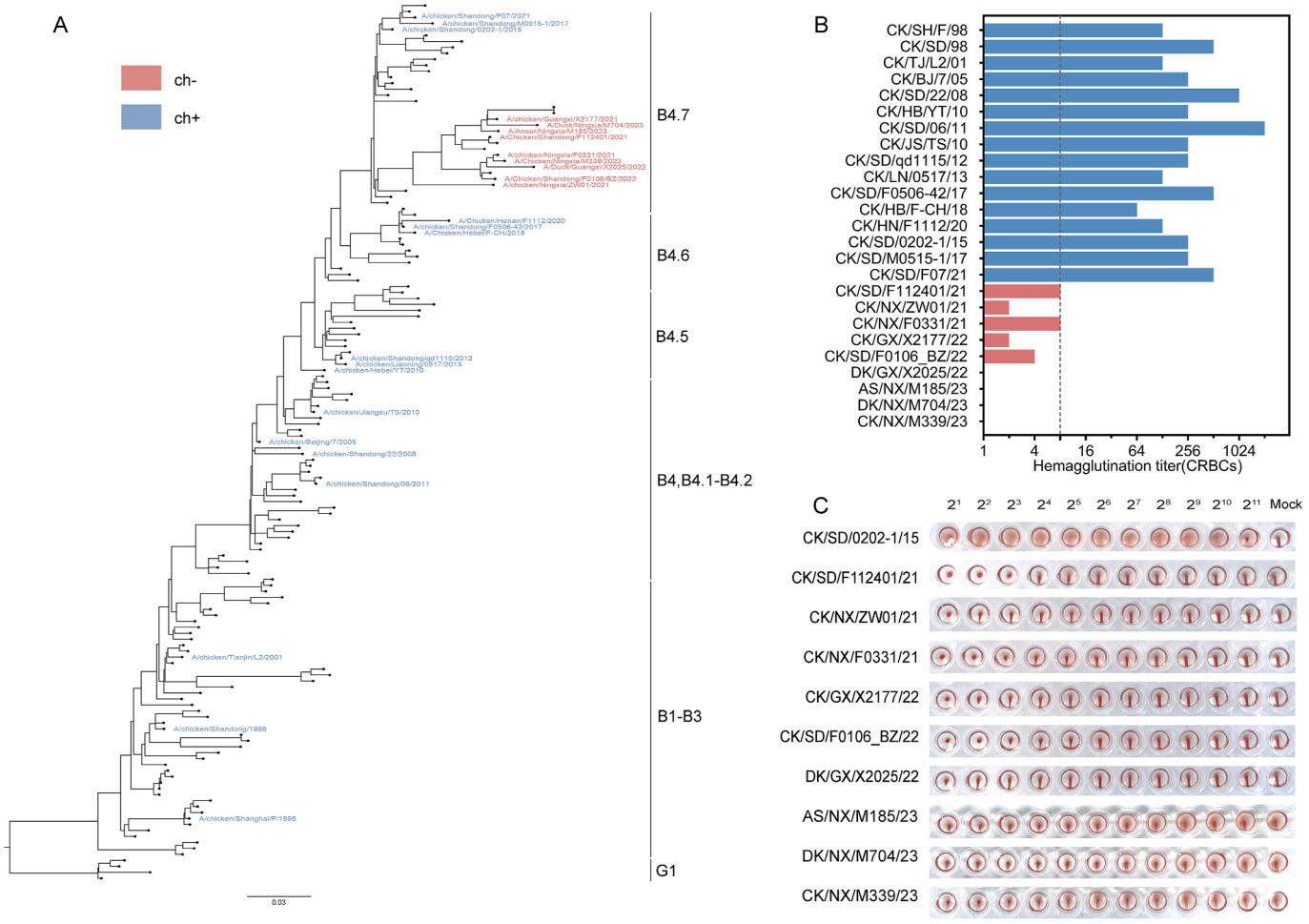


Fig. 1. Hemagglutination assays with chicken red blood cells (CRBCs) for H9N2 influenza viruses from different clades. **A** Phylogenetic tree of H9N2 *HA* genes (1998–2023), highlighting the 25 strains used in this study. Blue indicates strains with normal hemagglutination activity, whereas red indicates strains with reduced activity. **B**, **C** Hemagglutination titers of representative H9N2 strains from different years and clades using 1% CRBCs at 25 °C.

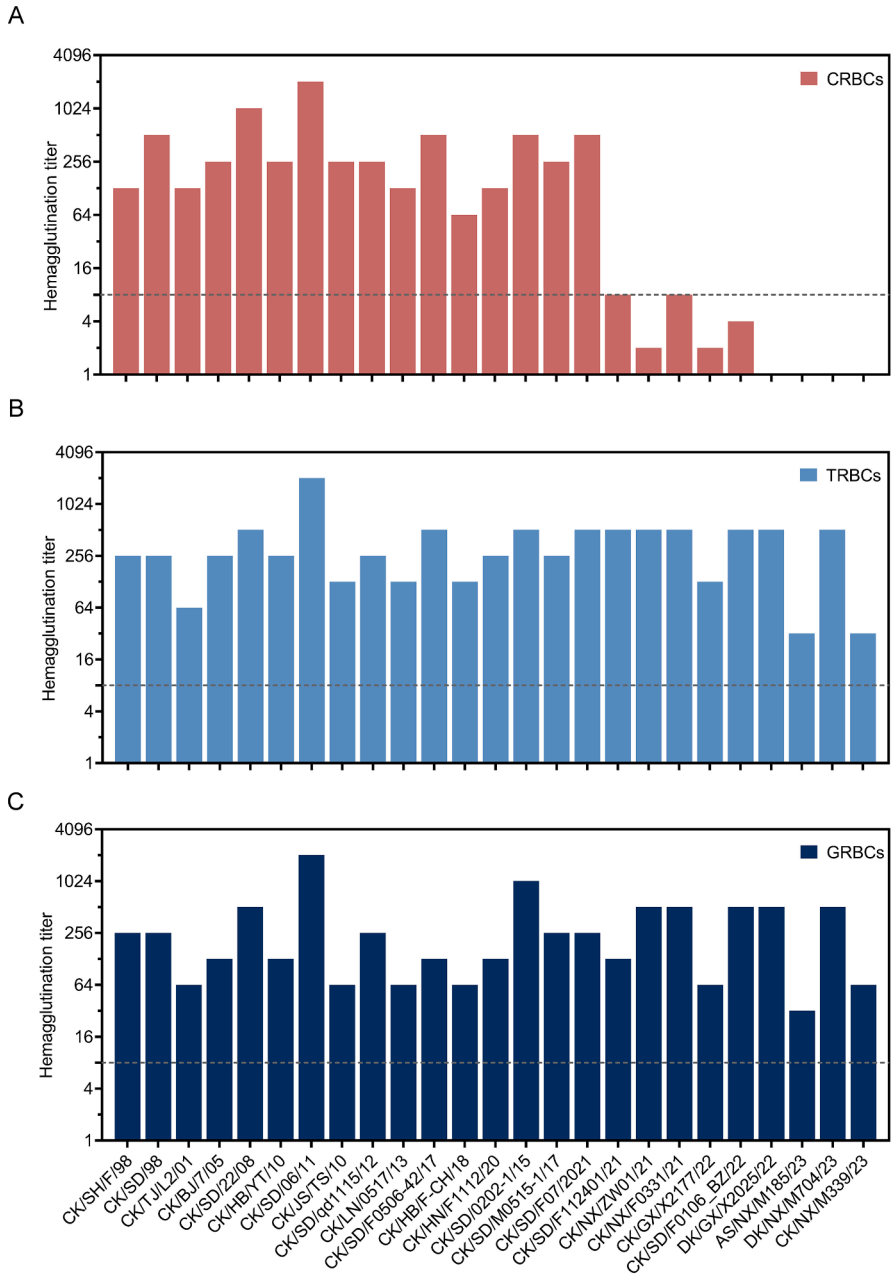


Fig. 2. H9N2 AIV strains with reduced CRBCs agglutination retain agglutination activity with turkey red blood cells (TRBCs) and guinea pig red blood cells (GRBCs). **A–C** Hemagglutination titers of H9N2 viruses measured using 1% CRBCs, TRBCs, and GRBCs at 25 °C.

maintained in our laboratory (1998–2023) showed that strains with reduced hemagglutination activity mainly clustered within the B4.7 lineage circulating during 2021–2023 (Fig. 1A; Supplementary Fig. S1). Hemagglutination assays at 25 °C further revealed that earlier strains (1998–2018) exhibited high titers, ranging from 64 to 2048 hemagglutination units (HAU), whereas most isolates from the B4.7 lineage (2021–2023) showed markedly lower titers, with many $\leq$ 2 HAU (Fig. 1B and C). Notably, all strains maintained viral titers of $\geq$ 6 $\log_{10}$ $TCID_{50}$/0.1 mL, excluding viral concentration as a confounding factor (Supplementary Table S2).

Following WHO recommendations, TRBCs and GRBCs were also used in hemagglutination assays as alternative indicators (World Health Organization, 2011, Chapter 2.E). Strikingly, all H9N2 AIV strains retained robust hemagglutination activity (32–2048 HAU) with TRBCs and GRBCs (Fig. 2A–C, Supplementary Fig. S2). These results indicate that since 2021, circulating H9N2 AIV strains have developed a specific decrease of CRBC agglutination ability while preserving efficient agglutination with TRBCs and GRBCs.

HA protein is responsible for reduced chicken RBC agglutination in H9N2 AIV

The agglutination of influenza viruses with red blood cells is primarily determined by the functions of hemagglutinin (HA) and neuraminidase (NA), where HA mediates binding to sialic acid receptors and NA facilitates receptor cleavage (Benton et al., 2015; Guo et al., 2018). To investigate the reduced CRBC agglutination observed in some H9N2 strains, we selected one representative strain with normal agglutination,

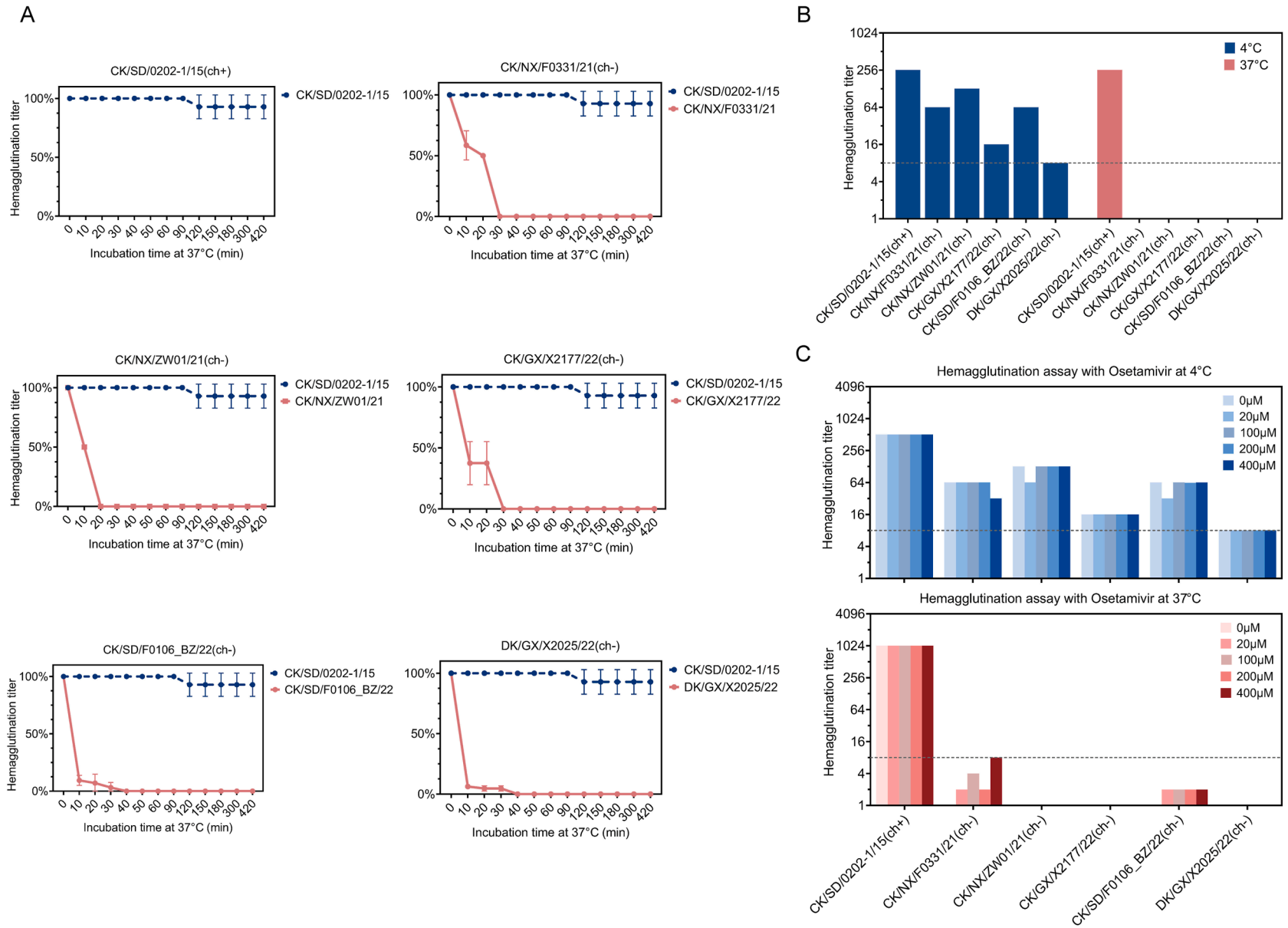


Fig. 3. Effects of HA and NA on hemagglutination activity of H9N2 AIV at 4 °C and 37 °C. **A** Virus elution assay using CRBCs. Viruses were adsorbed onto 1% CRBCs at 4 °C for 30 min and then incubated at 37 °C to monitor virus elution by measuring hemagglutination titers over 420 min. **B** Hemagglutination titers of representative H9N2 AIVs with CRBCs at 4 °C and 37 °C. **C** Hemagglutination titers with CRBCs at 4 °C and 37 °C after pre-incubation with increasing concentrations of oseltamivir carboxylate.

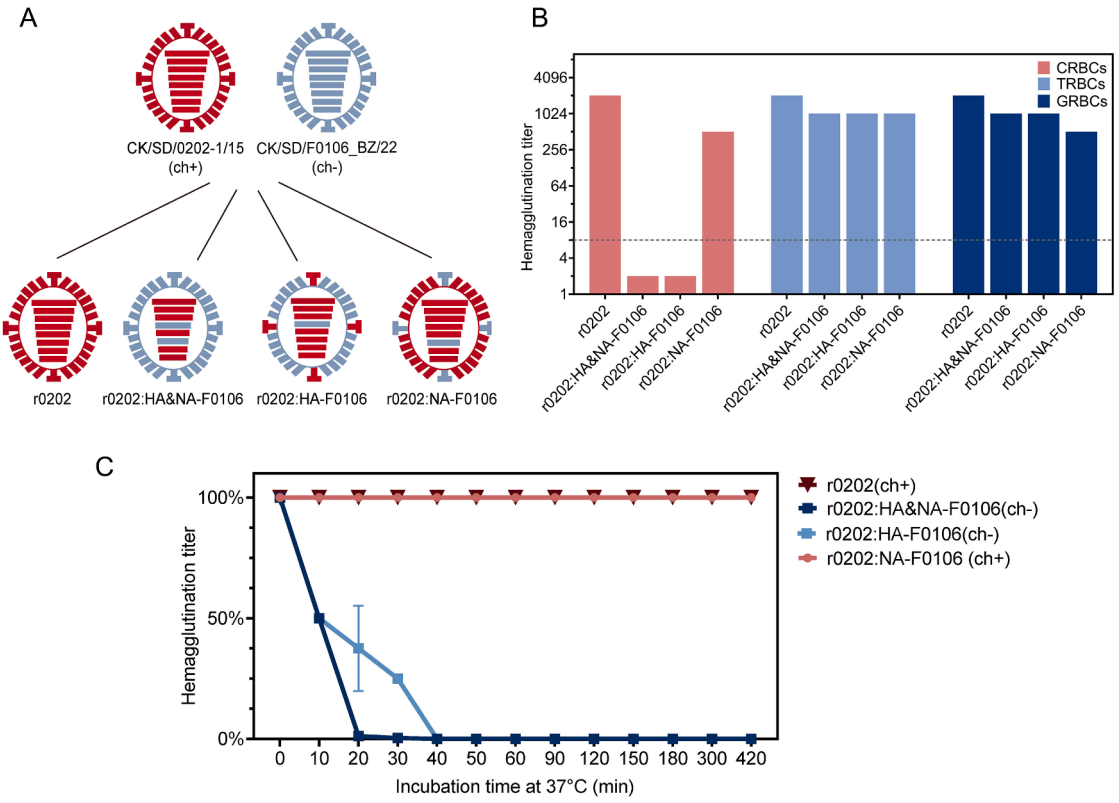


Fig. 4. Effects of HA and/or NA substitutions in recombinant H9N2 viruses on hemagglutination activity. **A** Rescue strategy for r0202, rF0106, r0202:HA-F0106, and r0202:NA-F0106. **B** Hemagglutination titers of the four recombinant viruses were measured with CRBCs, TRBCs, and GRBCs at 25 °C. **C** CRBC-based virus elution assays for the four recombinant viruses.

CK/SD/0202-1/15 (ch+), whose HA titer closely matched the infectious titer measured by $TCID_{50}$ in MDCK cells, and five strains exhibiting reduced agglutination (ch-) including CK/NX/F0331/21, CK/NX/ZW01/21, CK/GX/X2177/22, CK/SD/F0106_BZ/22 and DK/GX/X2025/22. Virus elution assays showed that ch+ viruses maintained hemagglutination titers for more than 420 min at 37 °C, whereas ch- viruses became undetectable within 30 min (Fig. 3A). Consistently, hemagglutination assays demonstrated that ch+ viruses agglutinated CRBCs efficiently at both 4 °C and 37 °C, while ch- viruses retained agglutination at 4 °C but displayed markedly reduced activity at 37 °C (Fig. 3B). In contrast, both virus groups showed comparable agglutination titers with TRBCs, indicating that the reduced CRBC agglutination phenotype results from intrinsic alterations in HA rather than temperature-dependent dissociation (Supplementary Fig. S3; Supplementary Table S3).

NA inhibition by oseltamivir (0–400 μM) produced minimal effects on CRBC agglutination (Fig. 3C; Supplementary Table S4), indicating that NA activity does not account for the phenotype. To further dissect the contributions of HA and NA, reverse genetics was used to generate recombinant viruses using the ch+ strain CK/SD/0202-1/15 as the backbone. Substitution of the *HA* gene from the ch- strain CK/SD/F0106_BZ/22 was sufficient to confer poor agglutination, whereas NA substitution had no detectable effect (Fig. 4B and C; Supplementary Table S5). Together, these data indicate that the reduced CRBC-agglutination phenotype is HA-dependent.

The 130-loop mutation in HA protein is responsible for the reduced hemagglutination activity of H9N2

The receptor-binding region of HA protein is essential for its attachment to sialic acid receptors on red blood cells (Medeiros et al., 2001). Comparing the HA sequence of ch+ (CK/SD/0202-1/15) and ch- (CK/SD/F0106_BZ/22) viruses revealed amino acid variations in the 130-loop (positions 131 and 132), 150-loop (positions 145 and 150), and 190-helix (positions 179, 180, and 183). Interestingly, all ch- viruses shared identical substitutions at positions HA motif 131/132-NT (Supplementary Table S6). Structural comparison showed that residue 131 N in a ch- virus (green) had a longer side chain projecting outward compared to 131T in a ch+ virus (pink), while residue 132T exhibited a rotated side-chain orientation (Fig. 5A, mature-HA numbering).

Using reverse genetics, we rescued the wild-type viruses r0202 (ch+) and rF0106 (ch-), as well as mutants targeting the 130-loop, 150-loop, or 190-helix. Hemagglutination assays showed that 130-loop mutations markedly reduced CRBC agglutination in r0202 (ch+) viruses, whereas the corresponding mutations increased CRBC agglutination in rF0106 (ch-) viruses. In contrast, mutations in the 150-loop or 190-helix caused only minor changes in hemagglutination titers, with an approximately twofold increase observed in some mutants (Fig. 5B; Supplementary Table S7). As expected, TRBC/GRBC agglutination remained intact across all mutants. Recombinant virus r0202:130 (HA motif 131/132-NT) exhibited markedly reduced CRBC agglutination (4 HAU) and lost activity within 60 min at 37 °C, whereas the reverse mutant rF0106:130 (HA motif 131/132-TA) restored agglutination (512 HAU) (Fig. 5B and C).

Hemadsorption assays in 293T cells further confirmed reduced CRBC binding for r0202:130 and enhanced binding for rF0106:130 (Fig. 5D; Supplementary Fig. S4). Together, these results identify the HA 130-loop as a key determinant of reduced CRBC agglutination, while the 150-loop and 190-helix variations are unlikely to be the main determinant of this phenotype.

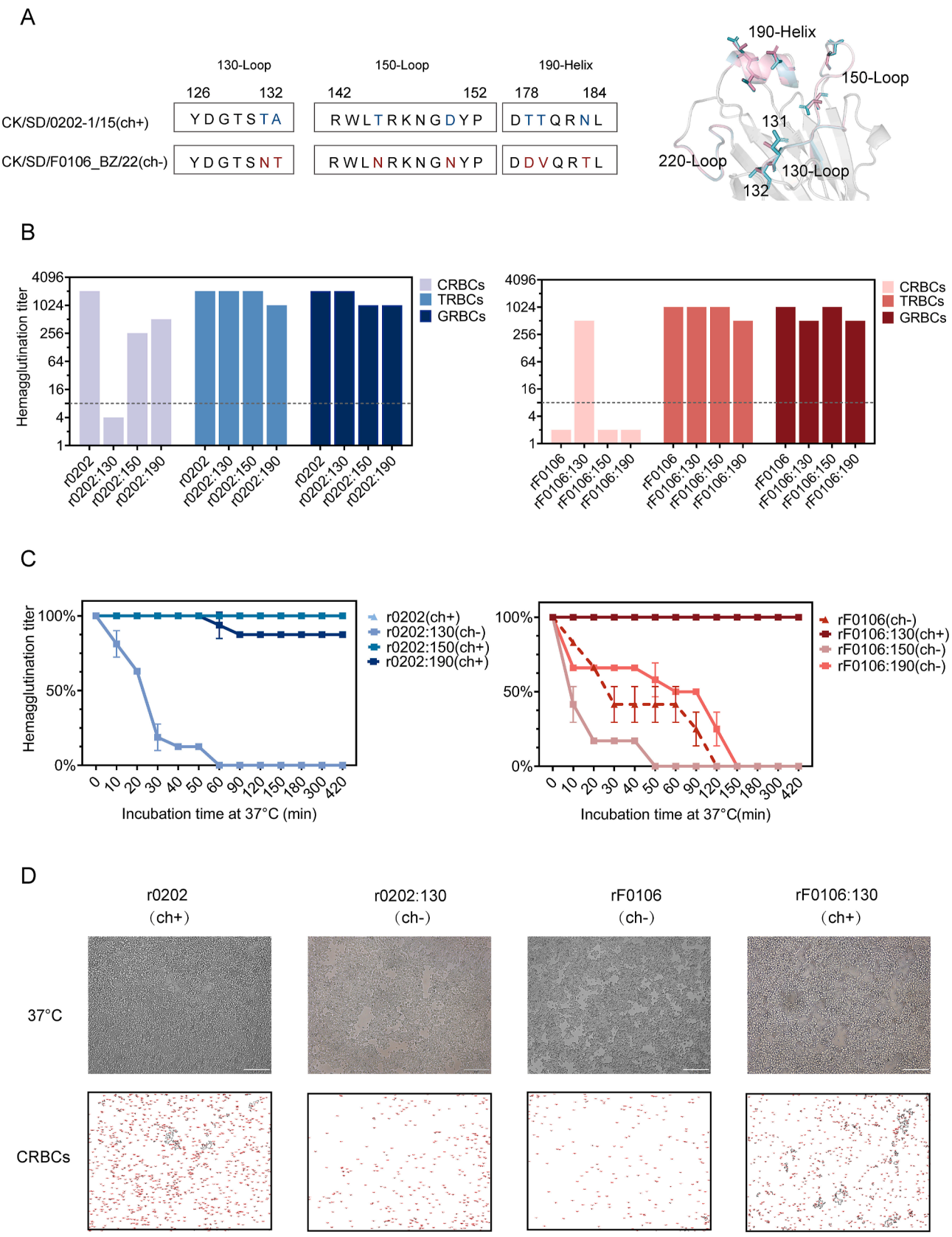


Fig. 5. The HA 130-loop mutation is responsible for reduced hemagglutination activity in H9N2 viruses. **A** Sequence alignment of the HA receptor-binding domain from CK/SD/0202-1/15 (pink) and CK/SD/F0106_BZ/22 (green), and structural analysis of amino acid differences in the HA protein. **B**, **C** Hemagglutination and virus elution assays with CRBCs for H9N2 mutant viruses. **D** Hemadsorption assay of H9 HA mutant proteins at the indicated temperatures. 293T cells expressing H9 HA mutants were incubated with 1% CRBCs, and adsorbed CRBCs attached to cells were visualized microscopically after washing (CRBCs; small, oval, nucleated cells attached to larger 293T cells). Lower panels show thresholded CRBC masks used for ImageJ-based quantification. Scale bar, 100 μm.

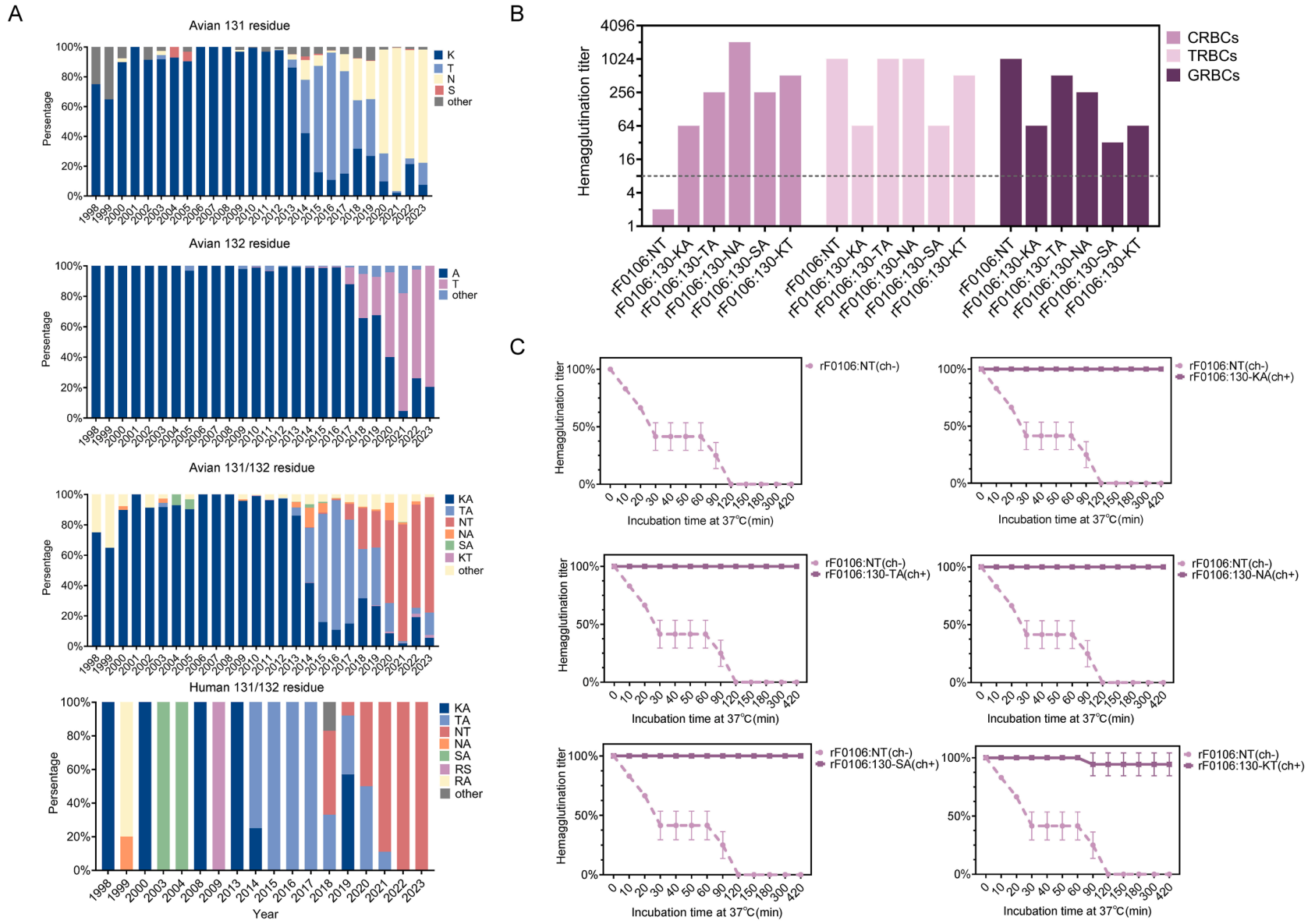


Fig. 6. Effects of the HA 131/132-NT motif on hemagglutination activity of H9N2 viruses. **A** Frequency of residues/motifs at HA positions 131 and 132 in avian and human H9N2 viruses from China between 1998 and 2023. A total of 8993 full-length HA sequences from China (1998–2023; GISAID database) were analyzed. **B** Hemagglutination titers of six H9N2 viruses with different HA 131/132 variants were measured using 1% CRBCs, TRBCs, and GRBCs at 25 °C. **C** Virus elution assays using CRBCs for the six HA 131/132 variant viruses.

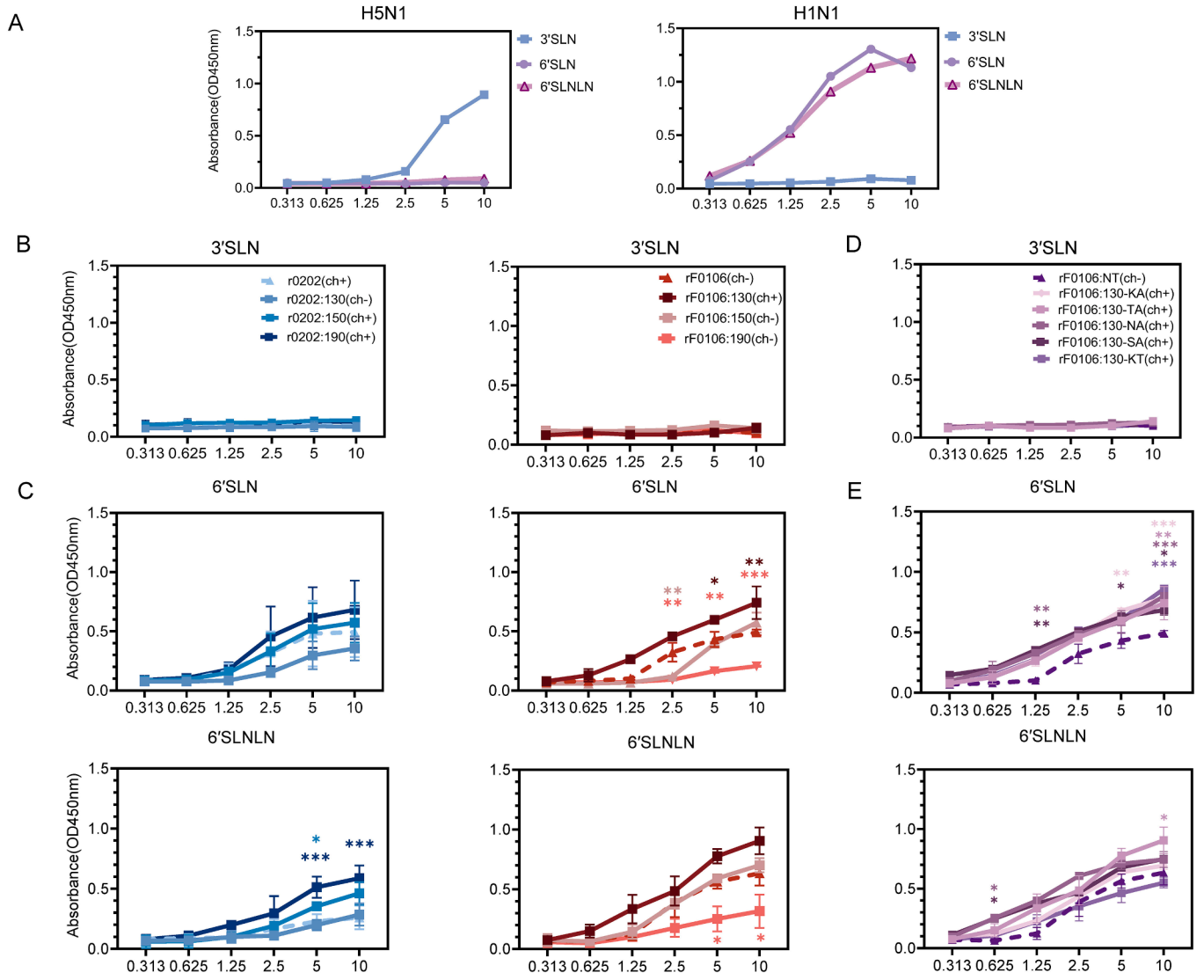


Fig. 7. Effects of the HA 131/132-NT motif on receptor-binding properties of H9N2 viruses. Binding affinities of (**A**) H5N1 and H1N1 control viruses and (**B**, **C**) H9N2 HA receptor-binding domain (RBD) mutant viruses and (**D**, **E**) HA 131/132 variant viruses to three glycopolymers were determined by solid-phase binding assay. Data are presented as mean ± SEM. Statistical significance was evaluated using two-way ANOVA (ns, not significant; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$).

HA motif 131/132-NT impairs CRBC agglutination of H9N2 avian influenza virus

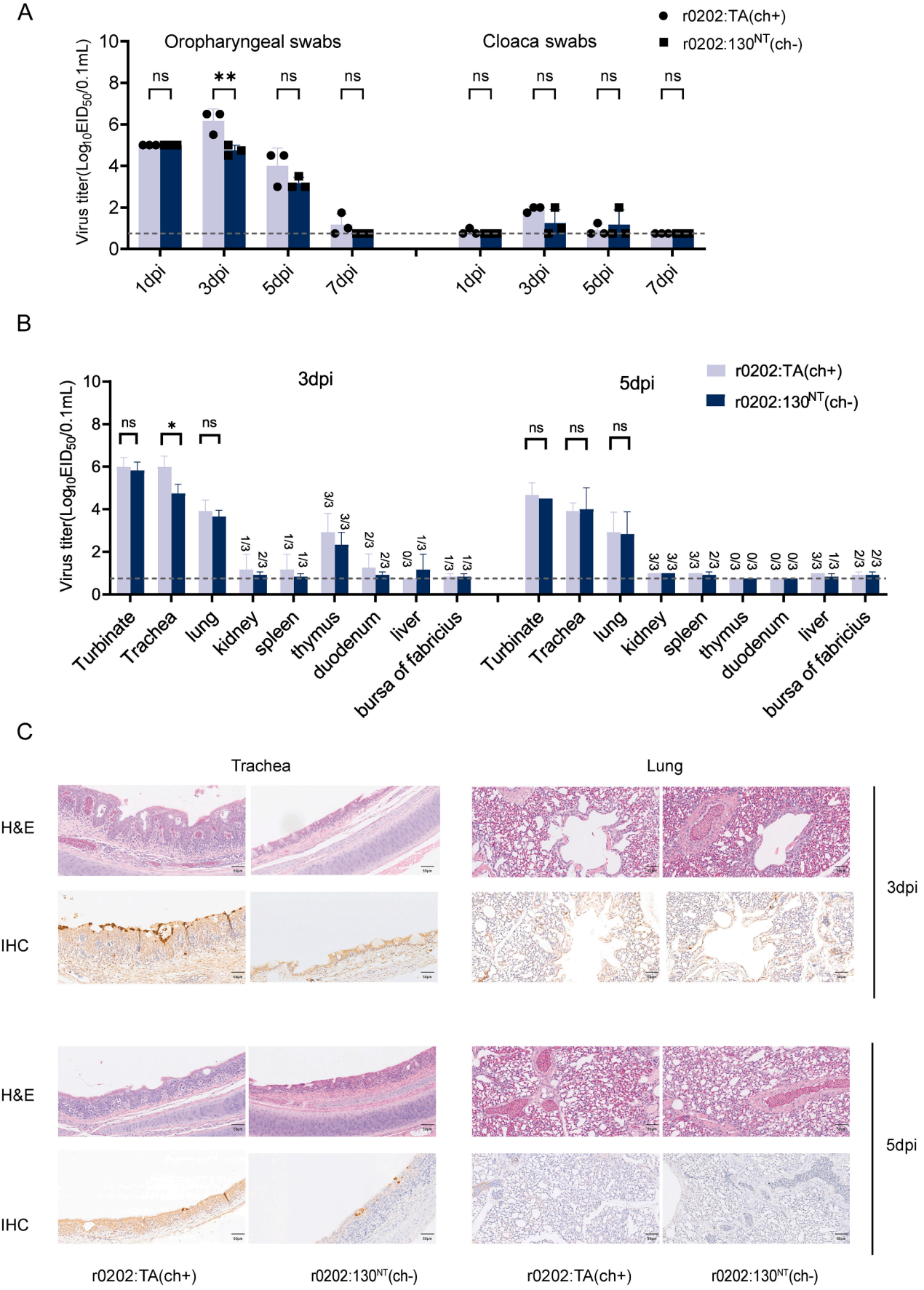
To investigate the evolutionary changes at HA positions 131 and 132, we analyzed avian H9N2 HA sequences from 1998 to 2023 (Supplementary Table S8). Both residues displayed substantial polymorphism. At position 131, lysine (K), threonine (T), asparagine (N), and serine (S) were detected, with T and K predominating before 2020 and N becoming dominant thereafter. At position 132, alanine (A) and threonine (T) were identified, with A being replaced by T after 2020. A concurrent shift in dominant residues at both positions was evident beginning in 2021 (Fig. 6A; Supplementary Fig. S5A).

Further analysis identified six 131/132 motifs among H9N2 viruses, of which KA, TA, and NT were the major epidemiologically relevant patterns, whereas the other motifs were rare. A similar trend occurred in human isolates: TA, KA, and NT comprised the majority of sequences, with NT becoming uniformly dominant after 2021 (Fig. 6A; Supplementary Fig. S5B). These parallel trends in poultry and humans suggest strong positive selection favoring the NT motif.

Based on these evolutionary observations, we constructed five mutant viruses—rF0106:130-KA, rF0106:130-TA, rF0106:130-NA, rF0106:130-SA, and rF0106:130-KT—using the rF0106 (NT motif) backbone. All five mutants restored CRBC agglutination and retained high hemagglutination titers at 37 °C, whereas rF0106 carrying the NT motif showed markedly reduced activity (Fig. 6B and C; Supplementary Table S9). These results demonstrate that the loss of CRBC agglutination is specifically attributable to the combined 131/132-NT motif, and that reversion of either residue is sufficient to restore function.

HA motif 131/132-NT reduces the binding of H9N2 virus to short-chain α2,6-linked sialic acid receptors

Hemagglutination activity of influenza viruses is closely related to their receptor-binding properties (Liu et al., 2023; Thompson et al., 2024). Since 2016, most H9N2 avian influenza virus isolates have primarily bound to α2,6 sialic acid receptors (Teng et al., 2016; Zou et al., 2019; Bi et al., 2020). To determine how the 131/132-NT motif alters



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receptor engagement, we examined binding of wild-type and mutant viruses to α2,3 (3′SLN) and short-chain α2,6 sialic acid receptors (6′SLN, 6′SLNLN). The H5N1 and H1N1 controls confirmed that the assay could detect α2,3- and α2,6-linked receptor binding, respectively (Fig. 7A). All viruses showed minimal binding to 3′SLN (Fig. 7B), but displayed clear differences in short-chain α2,6 binding (Fig. 7C). Within the r0202 backbone, the r0202:130 (ch-) mutant exhibited a decrease in binding to short-chain α2,6 sialic acid receptors, particularly to 6′SLNLN, while the r0202:150 (ch+) mutant showed a modest increase mainly in 6′SLNLN binding, and the r0202:190 (ch+) mutant demonstrated an increase in binding (Fig. 7C, left panels).

Conversely, within the rF0106 (ch-) backbone, the rF0106:130 (ch+) mutant showed increased binding to short-chain α2,6 glycans; the rF0106:150 (ch-) mutant caused only limited changes. By contrast, the rF0106:190 (ch-) mutant reduced short-chain α2,6 binding (Fig. 7C, right panels), indicating that substitutions outside the 130-loop can also modulate receptor binding. However, only the 130-loop substitution consistently correlated with the switch in CRBC agglutination phenotype in both viral backbones.

Consistently, rF0106 mutants in which NT was replaced by KA, TA, NA, SA, or KT motifs showed enhanced binding to short-chain α2,6 glycans, particularly to 6′SLN, while maintaining minimal binding to 3′SLN (Fig. 7D and E). These findings indicate that the HA motif 131/132-NT within the 130-loop reduces the binding of H9N2 AIV to short-chain α2,6 sialic acid receptors, which may contribute to their weakened ability to agglutinate CRBCs.

HA motif 131/132-NT maintains efficient replication of H9N2 viruses in chickens

To evaluate whether the reduced CRBC agglutination phenotype affects H9N2 virus replication *in vivo*, specific-pathogen-free (SPF) chickens were intranasally infected with 10^6 EID_{50}/0.2 mL of either r0202:TA (ch+) or r0202:130^{NT} (ch-). Following inoculation, both groups showed comparable oropharyngeal and cloacal shedding, with no significant differences at 1, 5, or 7 dpi. A transient reduction in oropharyngeal titers was observed for r0202:130^{NT} (ch-) at 3 dpi (Fig. 8A). At the same time point, viral loads of r0202:130^{NT} (ch-) in trachea tissues were temporarily lower than those of r0202:TA (ch+), but became comparable by 5 dpi. In contrast, viral loads in other examined organs, including the nasal turbinates, lungs, kidneys, spleen, thymus, duodenum, liver, and bursa of fabricius, showed no discernible differences between the two groups at either time point (Fig. 8B). Immunohistochemical analysis revealed reduced NP signals in the tracheas of r0202:130^{NT} (ch-)–infected chickens compared with r0202:TA (ch+) at 3 dpi, whereas staining in the lungs was comparable between groups. Histopathological examination showed that at 3 dpi, chickens infected with r0202:TA (ch+) exhibited moderate tracheal inflammation with vascular congestion, while those infected with r0202:130^{NT} (ch-) exhibited only mild and focal inflammatory infiltrates. By 5 dpi, tracheal lesions in both groups were mild and comparable. Pulmonary lesions were limited to occasional hemorrhage and congestion, with no evident differences between groups (Fig. 8C; Supplementary Fig. S6A). Collectively, these data indicate that viruses carrying the HA 131/132-NT motif retain efficient replication in chickens despite reduced hemagglutination activity.

HA motif 131/132-NT influences replication and tissue tropism of H9N2 viruses in mice

To assess the impact of the HA motif 131/132-NT on viral behavior in mammals, groups of 6-week-old BALB/c mice were intranasally inoculated with 10^6 $TCID_{50}$/50 μL of each virus. Throughout the 14-day observation period, no mortality or overt clinical symptoms were observed. Body weights remained largely stable in both groups, with no sustained weight loss observed (Fig. 9A and B). At 3 and 5 dpi, both viruses replicated efficiently in the nasal turbinate, showing comparable titers. However, r0202:130^{NT} (ch-) titers in the lungs were significantly lower than those of r0202:TA (ch+) and in some cases dropped below the limit of detection (Fig. 9C and D). Immunohistochemical analysis revealed comparable distributions of NP-positive cells in the nasal turbinates for both viruses, whereas r0202:TA (ch+) infection resulted in focal NP signals in the lungs, which were nearly absent in r0202:130^{NT} (ch-)-infected mice. Histopathological examination of the nasal turbinates revealed mild inflammatory infiltrates with limited epithelial alteration in both groups, without notable differences. In the lungs, both viruses induced focal mild inflammatory infiltrates at 3 dpi. By 5 dpi, r0202:TA (ch+) -infected mice still exhibited focal mild inflammation, whereas r0202:130^{NT} (ch-)-infected mice showed no appreciable lesions (Fig. 9E; Supplementary Fig. S6B). In summary, our findings suggest that the HA motif 131/132-NT alters the tissue tropism of H9N2 viruses in mice, with similar replication in the upper respiratory tract while attenuating viral growth in the lungs.

DISCUSSION

Our study reveals that the evolution of H9N2 avian influenza viruses (AIVs) has led to the emergence of variants with distinct hemagglutination phenotypes, revealing a subtle but significant shift in viral adaptation. The HA motif 131/132-NT is the key molecular signature driving this change. By reducing affinity for short-chain α2,6-linked sialic acid receptors, this motif likely underlies the weakened agglutination with CRBCs, reflecting an adjustment in receptor-binding preference rather than a loss of function. Notably, the 131/132-NT motif preserved viral infectivity in chickens while altering tissue tropism in mice, suggesting that this adaptation fine-tunes receptor engagement to support efficient infection in avian hosts and altered tissue tropism in mammals.

Influenza virus hemagglutination represents a finely tuned balance between HA-mediated receptor binding and NA-mediated receptor cleavage (Gaymard et al., 2016). Swapping the HA segments between the ch- and ch+ viruses reversed their CRBC agglutination phenotypes, indicating that the reduced agglutination in ch- viruses is primarily attributable to changes in HA. Consistent with this, the elution and NA-inhibition results in our study support reduced HA-binding avidity, rather than increased NA activity, as the main driver of the phenotype. However, because hemagglutination is determined by the functional balance between HA and NA, a contribution of HA–NA balance cannot be completely excluded. A recent study reported that an H9N2 strain (A/chicken/Baise/0701/2019) exhibited reduced hemagglutination activity, which was restored after successive passages in chicken embryos, and this recovery was linked to the N131K substitution in the HA protein (Wen et al., 2024). Notably, the HA 131/132-NT motif is essential for the reduced hemagglutination phenotype, as the single

Fig. 8. Chicken challenge study with recombinant H9N2 viruses. **A** Oropharyngeal and cloacal virus shedding at 1, 3, 5, and 7 dpi. Specific-pathogen-free (SPF) chickens ($n = 3$ per group) were inoculated with 10^6 EID_{50} of the indicated viruses. Virus titers were determined by EID_{50} assay (detection limit: 0.75 EID_{50}/0.1 mL). **B** Viral titers in organs at 3 and 5 dpi. Turbinates, trachea, lungs, kidneys, spleen, thymus, duodenum, liver, and bursa of Fabricius were titrated by EID_{50} assay ($n = 3$ per group per time point). **C** Representative histopathological findings in the respiratory tissues of infected chickens. Sections of trachea and lung were stained with H&E (upper) and influenza NP immunohistochemistry (lower). Scale bar, 50 μm. Statistical significance was evaluated using two-way ANOVA (ns, not significant; $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$).

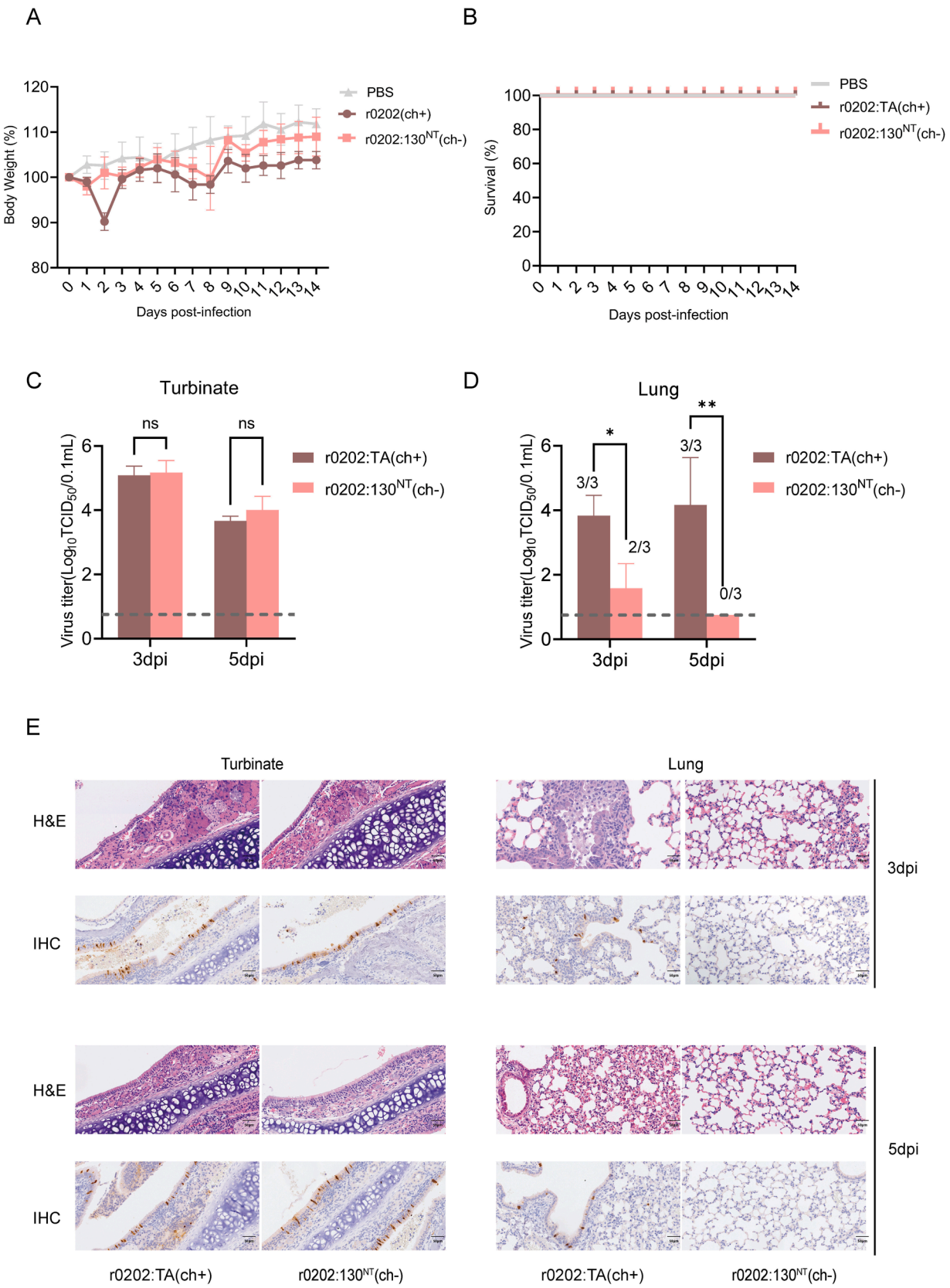


Fig. 9. Mouse challenge study with recombinant H9N2 viruses. **A**, **B** Groups of 13 mice were infected with the indicated virus at 10^6 $TCID_{50}$. Five mice per group were monitored for body weight and survival over 14 days. **C**, **D** Eight additional mice per group were euthanized at 3 dpi and 5 dpi, and virus titers in turbinates and lungs were determined by $TCID_{50}$ assay. **E** Representative histopathological findings in turbinates and lungs of infected mice. Sections were stained with H&E (upper) and influenza NP immunohistochemistry (lower). Scale bar, 50 μm. Data are presented as mean ± SD. Statistical significance was evaluated using two-way ANOVA (ns, not significant; $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$).

mutants rF0106:130-KT and rF0106:130-NA, compared with the parental rF0106 (NT) virus, still retained efficient hemagglutination, indicating that the phenotype arises from the combined action of residues 131 and 132.

Moreover, we found that mutations in the HA motif 131/132-NT not only impair hemagglutination with CRBCs but also reduce binding affinity for short-chain α2,6-linked sialic acid receptors (≤ 2 LacNAc repeats). Although CRBC agglutination is significantly reduced by T131 N/A132T, these variants still effectively agglutinate TRBCs and GRBCs. This difference likely arises from variations in receptor abundance and structural complexity on different RBCs. CRBCs are enriched in α2,3-linked receptors with relatively low levels of α2,6 glycans (Takemae et al., 2010; Broszeit et al., 2021), whereas TRBCs and GRBCs contain higher levels of α2,6. In addition, the presence of branched or extended glycan structures on these RBCs may further support hemagglutination (Byrd-Leotis et al., 2019; Broszeit et al., 2021).

Alterations in the receptor-binding properties of influenza viruses can markedly influence their infectivity and pathogenicity in host species (Ge and Wang, 2011). In chickens, consistent viral shedding and tissue titers showed that H9N2 viruses carrying the HA motif 131/132-NT retained robust infectivity. Previous studies have identified sulfated α2,3 receptors in chickens (Suzuki et al., 2022; Zhao and Pu, 2022), which are recognized by H9N2 viruses (Peacock et al., 2017). This interaction may account for their efficient infectivity in poultry, even with reduced binding to canonical α2,3 and α2,6 receptors. In mice, the HA motif 131/132-NT was associated with efficient replication in the nasal turbinates but limited replication in the lungs, consistent with receptor distribution in BALB/c mice, where both α2,3- and α2,6-linked sialic acids are present in the upper airways, while α2,3-linked receptors are more abundant in the lungs (Ning et al., 2009; Zhao and Pu, 2022).

Previous studies have shown that extended α2,6-sialylated N-glycans containing three or more LacNAc repeats are expressed on the respiratory epithelium of humans and ferrets (Suptawiwat et al., 2008; Jia et al., 2014). In the evolution of human seasonal H3N2 viruses, which originated from avian virus, receptor-binding specificity initially shifted from α2,3-linked to α2,6-linked sialic acids, followed by preferential binding to extended α2,6 glycans and loss of CRBC agglutination (Matrosovich et al., 2000; Nobusawa et al., 2000; Lin et al., 2012; Peng et al., 2017). During this adaptive process, human H3N2 viruses also showed attenuated pathogenicity in mice and limited binding to mouse respiratory tissues, with only rare binding to mouse alveolar cells (Van Riel et al., 2007; Lei et al., 2025). Similarly, H9N2 viruses have evolved from predominant binding to α2,3-linked receptors to increased affinity for α2,6-linked receptors (Wan and Perez, 2007; Zou et al., 2019; Bi et al., 2020). We here observed reduced binding to short-chain α2,6 glycans accompanied by diminished CRBC agglutination, suggesting that the 131 N/132T motif may follow the H3N2's receptor adaptation pathway.

The HA 131/132-NT motif has emerged as a dominant feature in both poultry and human H9N2 isolates, and our results demonstrate that it supports efficient infection in chickens while influencing tissue tropism in mammals, thereby implying a potential risk of interspecies adaptation. Its rapid predominance after 2021 may therefore involve additional selective factors beyond replication capacity in individual hosts. During the adaptation of H3N8, H7N9, and H9N2 viruses from waterfowl to chickens, a gradual shift toward α2,6-linked receptor recognition has been observed (Wan and Perez, 2007; Gambaryan et al., 2012; Tharakaraman et al., 2013; Sun et al., 2021; Thompson and Paulson, 2021; Yang et al., 2022; Cui et al., 2023; Wang et al., 2023). Although the 131/132-NT motif reduced binding to the short-chain α2,6 glycans tested here, this does not necessarily indicate a general loss of α2,6 receptor recognition. Instead, it may reflect altered HA preference toward extended or modified α2,6 glycans not examined in this study. Because α2,6 receptors are present in the chicken respiratory tract (Kuchipudi et al., 2009) but their detailed glycan composition remains unclear, this receptor-binding change may represent an adaptive feature in chicken populations. Interestingly, studies of engineered H7N9 viruses demonstrated that decreased affinity for short α2,6 glycans but enhanced binding to extended α2,6 glycans facilitated infection of both chicken and human respiratory epithelial cells (De Vries et al., 2017). Further investigation is needed to clarify how fine-tuned receptor-binding changes support efficient replication in poultry while shaping tissue tropism in mammals.

CONCLUSIONS

In summary, our research provides critical insights into H9N2 AIV evolution by identifying the 131 N/132T mutations as pivotal for hemagglutination and host adaptation. Therefore, we argue that continuous surveillance and refined detection strategies are imperative to safeguard poultry and track host shifts. Since the 131/132-NT variants can still agglutinate TRBCs and GRBCs, incorporating these assays into routine monitoring is essential to prevent oversight of emerging viruses.

MATERIALS AND METHODS

Viruses and cells

All H9N2 viruses used in this study were isolated during routine influenza surveillance in our laboratory (Supplementary Table S1) and propagated in 9–11-day-old SPF embryonated chicken eggs at 35 °C, then stored at −80 °C. Madin–Darby canine kidney (MDCK) cells and human embryonic kidney 293T (HEK293T) cells were cultured in DMEM (Gibco, New York, USA) supplemented with 10% FBS (Gibco, Sydney, Australia) and 1% penicillin–streptomycin (MACGENE, Beijing, China).

HA residues analysis and phylogenetic analysis

HA gene sequences were retrieved from the GISAID database (http://www.gisaid.org). A total of 8993 full-length HA sequences of influenza A viruses from China, including 8891 avian and 102 human isolates collected between 1998 and 2023, were analyzed. Sequences were aligned using MUSCLE v3.7, and maximum-likelihood phylogenetic trees were generated with IQ-TREE v1.6 under the GTR + I + G model with 1000 ultrafast bootstrap replicates. Phylogenetic trees were visualized and annotated using FigTree v1.4.4, Adobe Illustrator 2021, and Interactive Tree Of Life.

Plasmid construction and reverse genetics

As described previously (Sun et al., 2010), recombinant viruses were generated using a plasmid-based reverse genetics system in the genetic background of A/chicken/Shandong/0202-1/2015 and A/Chicken/-Shandong/F0106_BZ/2022. The desired mutations were introduced into the *HA* gene, and site-substitution viruses were rescued by reverse genetics in 293T cells and confirmed by RT-PCR and sequencing.

Hemagglutination assay

Hemagglutination assays were performed using 1% CRBCs, TRBCs, or GRBCs. Virus samples were twofold serially diluted in PBS in V (CRBCs and TRBCs)/U (GRBCs)-bottom plates and incubated with RBCs at 4, 25, or 37 °C for 30 min, and titers were recorded as HAU/50 μL. To evaluate the contribution of neuraminidase (NA) activity, the same assay was conducted in the presence of oseltamivir (0, 20, 100, 200, or 400 μM); virus dilutions were pre-incubated with the inhibitor for 60 min at room temperature before adding RBCs at 4 °C or 37 °C.

Hemagglutination assay with NA inhibitor

Viruses were twofold serially diluted in PBS with 0, 20, 100, 200, or 400 μM oseltamivir (MedChemExpress, USA), incubated at room temperature for 60 min, and assessed for hemagglutination with 1% CRBCs or TRBCs at 4 °C or 37 °C.

Hemadsorption assay

293T cells were transfected with pCAGGS-HA expression plasmids using jetPRIME (Polyplus, France). At 4 h post-transfection, the medium was replaced with DMEM containing 10% FBS, and cells were incubated for an additional 24 h. Cells were then incubated with 500 μL of 1% CRBCs at 37 °C for 30 min. After four washes with PBS, CRBC adsorption was visualized by light microscopy and quantified using ImageJ.

Virus elution assay

H9N2 influenza viruses were adsorbed to 1% CRBCs at 4 °C for 30 min, and the HA titers at 37°C—representing viral elution from CRBCs—were monitored at the indicated time points up to 420 min. The HA titers following incubation at 37 °C were recorded as the percentage of the HA titers at time zero at 4 °C.

Receptor-binding specificity assay

Receptor binding specificity was analyzed with a solid-phase binding assay as previously described (Matrosovich and Gambaryan, 2012). Briefly, plates (Thermo Fisher 15500) were coated overnight at 4 °C with serial dilutions of α2,3 SA (3′SLN: Neu5Acα2-3Galβ1-4Glcβ-sp4-PAA-biotin), α2,6SA (6′SLN: Neu5Acα2-6Galβ1-4GlcNAcβ-sp3-PAA-biotin), or 6′SLNLN (Neu5Acα2-6Galβ1-4GlcNAcβ-4GlcNAcβ-sp3-PAA-biotin) (GlycoNZ). Viruses were standardized to 64 HAU (using TRBCs) and incubated on blocked, PBST-washed plates for 10 h at 4 °C. Bound virus was detected using chicken anti-H9N2 serum followed by HRP-conjugated anti-chicken antibody (1:10,000). Color was developed with TMB for 10 min, stopped with 0.5 M H_2SO_4, and absorbance was read at 450 nm.

Chicken challenge studies

Six-week-old SPF chickens (n = 9 per group) were intranasally inoculated with 10^6 EID_{50} of each virus in 200 μL. Three birds per group were euthanized at 3 dpi and 5 dpi for viral titration in collected tissues (detection limit: 0.75 EID_{50}/0.1 mL). Trachea and lung tissues collected at these time points were fixed in formalin, paraffin-embedded, sectioned, and examined by H&E and NP immunohistochemistry. The remaining three birds were monitored, and oropharyngeal and cloacal swabs were collected at 1, 3, 5, and 7 dpi for virus titration.

Mouse challenge studies

Groups of thirteen 6-week-old female BALB/c mice were anesthetized with Zoletil 50 (20 mg/kg, intramuscular) and inoculated intranasally with 10^6 $TCID_{50}$ of virus or PBS in 50 μL. At 3 and 5 dpi, four mice per group were euthanized, and nasal turbinates and lungs were collected for viral titration in MDCK cells and were fixed in formalin, paraffin-embedded, sectioned, and examined by H&E and NP immunohistochemistry. The remaining five mice were monitored daily for 14 days for clinical signs, body weight changes, and mortality; animals losing > 25% of their initial body weight were euthanized.

Statistical analysis

Graphing and statistical analyses were performed using GraphPad Prism 8. Data are presented as mean ± SD or SEM and were analyzed by two-way ANOVA (ns, not significant; $*P < 0.05$; $**P < 0.01$; $***P < 0.001$).

DATA AVAILABILITY

The genome reference sequence of H9N2 viruses generated in this study have been deposited in the NCBI GenBank database. All HA sequences for this article are available on the Science Data Bank (https://doi.org/10.57760/sciencedb.v.00018). Isolate names and corresponding accession numbers are listed in Supplementary Table S1. All primary data supporting the findings of this study are available within the article and its Supplementary Materials.

ETHICS STATEMENT

9–11-day-old specific pathogen-free (SPF) embryonated chicken eggs and 6-week-old SPF female chickens were purchased from Boehringer Ingelheim, China. Six-week-old SPF female BALB/c mice were purchased from Vital River Laboratory Animal Technology Co., Ltd. (Beijing, China). All animal research was approved by the Beijing Association for Science and Technology (SYXK, Beijing, 2018-0038) and conducted in compliance with the Beijing Laboratory Animal Welfare and Ethics guidelines and the CAU Institutional Animal Care and Use Committee guidelines (SKLAB-B-2010-003). All experiments involving H9N2 viruses were performed under biosafety level 2 (BSL-2) containment conditions in accordance with institutional biosafety requirements.

AUTHOR CONTRIBUTIONS

Xue Chen: conceptualization, methodology, investigation, formal analysis, writing-original draft, writing-review & editing; Jinze Dong: investigation, validation, formal analysis, writing-review & editing; Chuankuo Zhao: software, investigation, formal analysis, writing-review & editing; Linlin Wang, Xinyang Gao and Zhixin Li: data curation, visualization; Honglei Sun, Yipeng Sun and Jinhua Liu: resources, supervision; Rui Zhang: conceptualization, supervision, writing-review & editing; Juan Pu: conceptualization, supervision, project administration, writing-review & editing, funding acquisition.

CONFLICT OF INTEREST

The authors declare that they have no competing interests.

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APPENDIX A. SUPPLEMENTARY DATA

Supplementary data to this article can be found online at https://doi.org/10.1016/j.virs.2026.06.007.

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